ASHISH KUMAR

Senior AI/ML Engineer · GenAI & LLM Systems · Multi-Agent AI · MLOps · Healthcare AI

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SUMMARY

Senior AI/ML Engineer with 3+ years building production GenAI systems, multi-agent architectures, and scalable ML pipelines.

Specialized in Agentic RAG, real-time LLM inference, RLHF-aligned system design, and end-to-end ML lifecycle across healthcare and cloud domains.

Delivered **$2.1M+ cost savings**, **96% latency reduction (18 min → 38 sec)**, and **25% reduction in clinical workload**.

CORE SKILLS

GenAI & LLM	GPT-4o, LangChain, LangGraph, CrewAI, Agentic RAG, RLHF, Reward Modeling, Prompt Engineering, Function Calling, Hugging Face, LoRA/PEFT
ML Engineering	Scikit-learn, Time-Series Forecasting, Feature Engineering, Model Evaluation at Scale (F1/P/R/AUC), Hyperparameter Tuning, Fine-tuning
MLOps & Deploy	FastAPI, MLflow, Docker, Kubernetes, GitHub Actions, CI/CD, A/B Testing, Canary Deploys, Drift Detection, LangSmith
Data & Pipelines	PySpark, Databricks (Delta Lake, DLT), SQL, ETL, Batch + Streaming, Azure ML Pipelines, Pinecone
Cloud	Azure (OpenAI, ML, AI Search, Functions, Fabric) · AWS (SageMaker, Lambda, S3) · GCP (Cloud Run)
Healthcare AI	FHIR R4/R5, HL7 v2/v3, HIPAA Compliance, EHR Integration, Clinical NLP

EXPERIENCE

Senior AI/ML Engineer *Apr 2025 – Present*

MEDISYNCAI INC. · Remote (San Francisco HQ)

- Architected 7-agent Agentic RAG system for clinical workflows (LangChain, LangGraph, Azure OpenAI) → **physician review time -25%, early detection +42%**
- Built real-time LLM voice agent (FastAPI, WebSockets, OpenAI Realtime API, Twilio) → **<200ms latency, nurse workload -30%**
- Deployed 7B-parameter fine-tuned clinical LLM (LoRA/PEFT on Azure ML) serving 1,000+ patient records/day → **500+ concurrent users, 99.95% uptime**
- Designed FHIR R4/R5 + HL7 EHR integration pipelines across 5 hospital systems → **100% data fidelity, zero HIPAA audit findings**
- Designed distributed LLM systems with streaming inference and fault-tolerant pipelines → **40% query speedup, sub-second clinical decision support**
- Built model evaluation framework at scale (F1 ≥ 0.92, AUC, RLHF preference scoring) embedded in CI/CD → **automated quality gates on every deployment**
- Mentored 4 engineers on LLM system design, RLHF workflows, and production ML best practices

AI/ML Engineer *Jan 2023 – Mar 2025*

THE UNIFIED CLOUD · Noida, India

- Built time-series forecasting system (lag features, rolling stats, Scikit-learn on Databricks) → **$2.1M annual savings, 35% cost reduction across 15K+ nodes**
- Engineered end-to-end ML pipeline (Delta Lake → **PySpark -> Azure ML -> FastAPI) with canary deploys, auto-retraining, and drift detection**
- Designed multi-agent IT automation system (LangChain, GPT-4, tool use) → **4,800+ ops hours saved/year, MTTR -60%**
- Led migration of 12 legacy systems to cloud-native microservices → **release velocity 3×, failure rate -65%**
- Led and mentored team of 6 engineers across ML, data pipelines, and platform infrastructure

OPEN SOURCE & PUBLICATIONS

Agentic-RAG-Clinical ·
Open-source 7-agent clinical orchestration framework (LangGraph + FHIR). 200+ GitHub stars. Used in production healthcare AI deployments.

Technical Article · "How We Reduced Clinical LLM Latency by 96%" · Medium / Towards Data Science
Deep-dive on distributed inference architecture, streaming pipelines, and fault-tolerant multi-agent design. 5,000+ reads.

KEY PROJECTS

Agentic Clinical Orchestrator
Latency 18 min → 38 sec · F1: 94.2% · Early detection +42% · HIPAA-compliant · 200+ GitHub stars
7-agent LangGraph system (triage → risk scoring → EHR write-back) on Azure ML; Databricks streaming pipeline for real-time clinical inference.

Enterprise RAG System — 4.8 TB
Retrieval 25 min → 8 sec · 99.3% citation accuracy · 4.8 TB indexed
PySpark + Databricks Delta Lake ingestion into Azure AI Search + Pinecone; LangChain hybrid retrieval with reranking; LangSmith observability.

Cloud Cost Intelligence Engine
$2.1M savings · <4% MAE · 15,000+ nodes · 41% spend reduction
Databricks time-series forecasting (Scikit-learn + PySpark) automated Reserved Instance purchasing; deployed via FastAPI with Azure ML model registry.

EDUCATION

Gurukula Kangri (Deemed University), Haridwar, India
MCA — Computer Science · GPA 8.65 · 2021–2023
BSc — Computer Science · GPA 8.18 · 2018–2021

CERTIFICATIONS

- AWS Bedrock: Generative AI Foundations (2024)
- Machine Learning & Deep Learning Specialization
- Data Structures & Algorithms in C++